

Timothy Do

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EDUCATION

- University of California, Los Angeles** Sep. 2023 – Dec. 2024 (*Expected*)
M.S. Electrical and Computer Engineering, Signal Processing Specialization Los Angeles, CA
- **Relevant Coursework:** Computational Imaging, Digital Speech Processing, Deep Learning & Neural Networks
 - **Clubs and Societies:** ECEGAPS (Electrical and Computer Engineering Graduate and PostDoc Society)
- University of California, Irvine** Sep. 2019 – June 2023
B.S. Electrical Engineering, Accelerated Status Irvine, CA
- **Cumulative GPA:** 3.924/4.0, **Major GPA:** 3.944/4.0, **Magna Cum Laude**
 - **Specializations:** Digital Signal Processing, Communications
 - **Relevant Coursework:** Computer Architecture, Control Systems, Digital Image/Signal Processing, Electronics
 - **Clubs and Societies:** Irvine Computer Vision Laboratory, IEEE, IPC, Eta Kappa Nu, Tau Beta Pi

EXPERIENCE

- Embedded Software Engineering Intern** June 2023 – Present
Maxon Solar Technologies San Jose, CA
- Implemented Robust OpenCV pipelines to quantitatively detect module features for defect binning
 - Developed Tkinter GUI interfaces for operator ease of use when testing images under custom data pipelines
 - Performed Multi-Class Data Labeling and Implemented Data Augmentation methods for training robust YOLOv5 Defect Classification Models
- Lead Undergraduate Student Researcher** January 2022 – June 2023
Irvine Computer Vision Laboratory at UC Irvine Irvine, CA
- Researched under Professor Glenn Healey on the topic of Spectral Image Filtering and Baseball Simulations
 - Digitized Hyperspectral Optical Imaging Filters into CSVs and compared them against certain chemical emissions
 - Managed 15+ Graduate/Undergraduate Researchers in Google Spaces and developed shared MATLAB/Python Scripts hosted on Google Drive/Github for analyzing SWIR/NVIR vegetation spectras
- Industrial Internet of Things Intern** June 2022 – Sep. 2022
Western Digital San Jose, CA
- Applied Industrial Toolsets to develop a Raspberry Pi Sensor Data Analytics Platform
 - Earned the Six Sigma Yellow Belt for learning foundations of Lean Six Sigma
 - Saved the company an annual projected cost of \$312,000 from hazard mitigations

PROJECTS

- RuHuman Audio Verification System** | *Python, MATLAB, Javascript, Librosa, Flask* Sep. 2023 – Dec. 2023
- Developed a resilient audio verification system to classify between bonafide and spoofed speech
 - Investigated the effects of AWGN on the ASVspoof2021 set based on t-DCF & EER metrics
 - Implemented a Flask Server s record/upload audio samples for verification from any web device with a microphone
- The Do-Pro Senior Design Project** | *Raspberry Pi, Python, MATLAB, OpenCV* Sep. 2022 - March 2023
- Led a team of four in creating an embedded stereo-vision camera for ADAS detection and 3D-Reconstruction
 - Calibrated the stereo-camera in MATLAB and implemented block matching algorithms with OpenCV
 - Designed the button/touch-screen user interface using RPi.GPIO and Tkinter for navigating the camera
 - Developed a Power Regression Model for the Do-Pro to percieve depth with 6% error from 1-3 feet
- EECS195-Colorization** | *Python, MATLAB, OpenCV, Tensorflow* Feb. 2022 – March 2022
- Colorized grayscale images using GAN and Auto-Encoder Neural Networks with finetuning on custom data
 - Gathered custom datasets by taking videos of different scenery with distinct objects then split them into frames
 - Clustered pixels into superpixel subgraphs for uniform training weights during preprocessing

SKILLS

Electronics: Raspberry Pi, Network Analysis, Cadence, LTSpice, Soldering, Oscilloscope & Multimeter Measurements
Programming Languages: Python, Java, C, CUDA, MATLAB, bash, MySQL HTML/CSS/Javascript, VHDL
Developer Tools: git, SSH, Makefile, Linux, Azure, VS Code, Visual Studio, PyCharm, Eclipse, Vivado HLS
Libraries: OpenCV, Pandas, Numpy, Matplotlib, PIL, Tensorflow, Pytorch, Tkinter, Socket, Flask, Librosa